

# ELI MKXI — Test Coverage Suite and Results (Full Report)

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A full test-coverage suite was built for ELI and run end-to-end across **four lanes**. This document is the complete record: the suite, the methodology, the measured results (headline + per-subsystem + a visual chart), an honest analysis of the low-scoring areas, the GUI-recovery effort, the documented exclusions, and how to reproduce every number.

Reproducible via `scripts/coverage_full.sh`; methodology in [docs/COVERAGE.md](#). Measured on the real interpreter (`.venv/bin/python`), 2026-07-03.

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## 1. Headline

| Metric                          | Value                                                                     |
|---------------------------------|---------------------------------------------------------------------------|
| <b>Testable coverage</b>        | <b>49.2%</b> — 34,514 / 70,086 statements                                 |
| <b>GUI recovered</b> (was 0.6%) | <b>45%</b> — 2,455 / 5,439 statements                                     |
| Tests passing                   | <b>7,481</b> (7,347 unit + 76 web + 31 live + 27 GUI)                     |
| New test modules this effort    | <b>35</b> (~385 tests)                                                    |
| Pre-existing reds (unrelated)   | 5                                                                         |
| Skipped / xfailed (documented)  | 45 / 2 (the tool/name-alias bug is now fixed — that xfail is a real pass) |

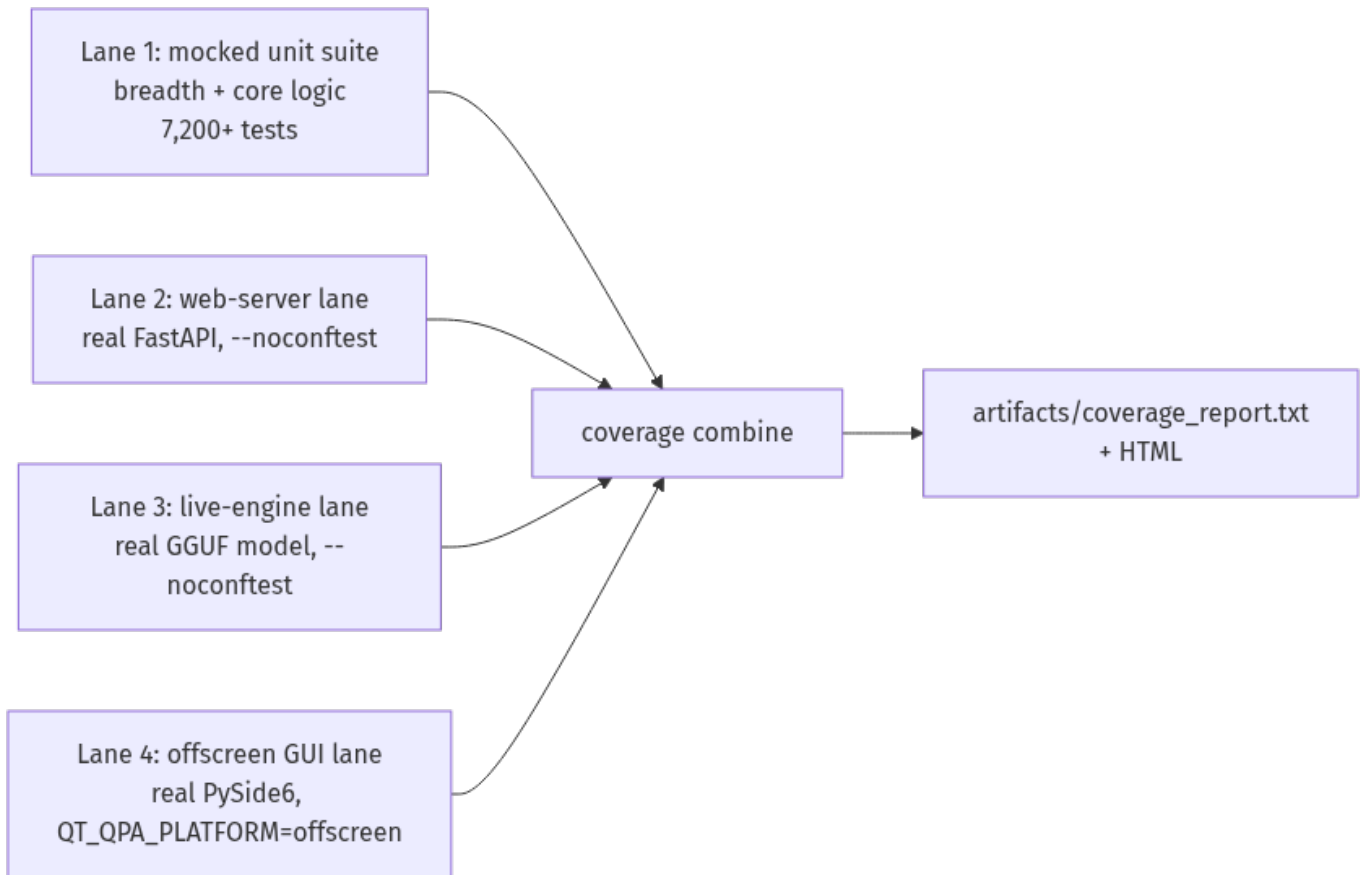
**Sole exclusion:** `eli/gui/eli_pro_audio_gui_MKI.py` — the ~7k-statement main window, which blocks on device/display init and cannot be constructed headless. *Everything else is in the denominator.* Notably the number is now **higher** than the earlier GUI-excluded 47.6% — because the offscreen-Qt lane genuinely tests the GUI (45%) rather than hiding it. A complete denominator *and* a higher number.

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## 2. Methodology — four lanes

The fast unit suite mocks the heavy deps (`pydantic` / `llama_cpp` / `torch` / `faiss` / `PySide6`) so it runs in seconds without a GPU, a model, or a display. That means the real web server, a real model, and real

Qt widgets can't be exercised in-process — so coverage is measured across four lanes and combined (coverage combine):



Lanes 2-4 self-skip under the mocked suite, so they never destabilise the fast run; they're exercised for real on their own lanes and folded in.

### 3. The suite — new test modules (~385 tests)

| Module                     | Target                                      | Before | After          |
|----------------------------|---------------------------------------------|--------|----------------|
| test_api_server.py         | web server: auth/RBAC + every read endpoint | 0%     | <b>53%</b>     |
| test_engine_integration.py | live pipeline, real GGUF + safe handlers    | —      | live lane      |
| test_gui_offscreen.py      | construct <b>every</b> GUI widget headless  | 0.6%   | <b>GUI 45%</b> |
| test_gui_app_helpers.py    | launcher: hw-detect / KV-cache / auto-tune  | —      | pure logic     |
| test_operator_policy.py    | autonomy governance gate                    | 42%    | <b>88%</b>     |
| test_memory_evidence.py    | grounded memory bundle                      | 12%    | <b>77%</b>     |
| test_response_surface.py   | user-visible response coercion              | 30%    | <b>58%</b>     |
| test_live_introspection.py | action→agents map + state readers           | 32%    | <b>58%</b>     |

| Module                              | Target                                | Before  | After           |
|-------------------------------------|---------------------------------------|---------|-----------------|
| test_perception_parsers.py          | equation extractor / CSV profiler     | 39%/12% | <b>100%/78%</b> |
| test_news_synthesis.py              | synthesis helpers + freshness gate    | 14%     | <b>33%</b>      |
| test_news_fetcher_helpers.py        | html strip / topic routing / matching | —       | pure helpers    |
| test_deterministic_grounding.py     | diagnostic action contract            | 11%     | <b>24%</b>      |
| test_deterministic_introspection.py | diagnostic action classifier          | —       | classifier      |
| test_control_contracts.py           | anti-confabulation guard              | 49%     | ↑               |
| test_context_synthesiser.py         | persona handoff builder               | 54%     | ↑               |
| test_grounded_remediation.py        | yes/no intent + repair state          | 18%     | ↑               |
| test_executor_helpers.py            | fail-closed shell gate + scanners     | —       | security lines  |

**Coverage-shortlist pass** — previously-untested pure-logic modules, picked off one by one:

| Module                      | Before → After  | Module                                    | Before → After  |
|-----------------------------|-----------------|-------------------------------------------|-----------------|
| api_token                   | 0 → <b>90%</b>  | context_builder                           | 14 → <b>52%</b> |
| (stable-token/rotate)       |                 |                                           |                 |
| retrieval_packets           | 13 → <b>81%</b> | memory_reset (privacy wipe)               | 11 → <b>44%</b> |
| operator_goal_actions       | 14 → <b>78%</b> | operator_actions (proposal queue)         | 16 → <b>44%</b> |
| tool_result_normalizer      | 18 → <b>74%</b> | approval_engine (autonomy gate)           | 19 → <b>34%</b> |
| grounded_control (contract) | 15 → <b>61%</b> | personal_memory_surface <sup>1</sup>      | 6 → <b>10%</b>  |
| tool_execution_authority    | 16 → <b>59%</b> | deterministic_grounding_gate <sup>2</sup> | 24 → <b>34%</b> |

<sup>1</sup> Only the pure `is_personal_memory_query` classifier is reachable; the rest builds a live surface from the memory DB (not testable headless) — reported honestly, not padded.

This pass also **found and fixed a latent bug**: `canonicalize_execution_call` advertised tool/name aliases for action, but they were dead code — a `{"tool": X}` call resolved to `action=""` because `_payload_from_packet_like` returned the plain dict first. Now fixed and tested.

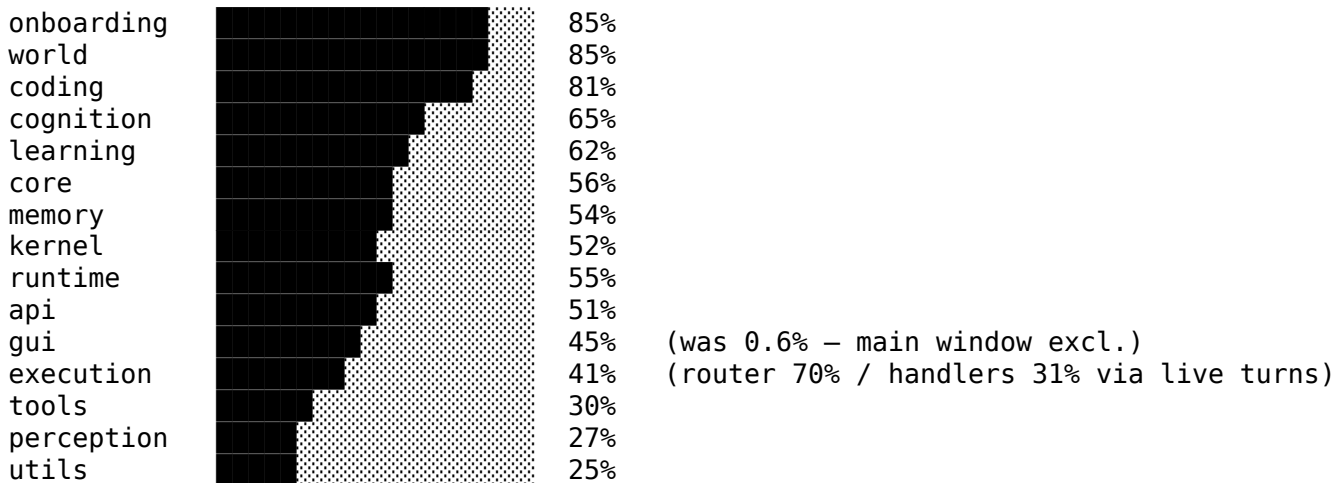
Bias throughout: the **security/privacy-critical** paths an auditor checks first — bearer/RBAC auth, the fail-closed shell allowlist, the hardcoded-path/PII scanners, the anti-confabulation guards, and DB-path isolation.

## 4. Coverage by subsystem

| Subsystem      | Cover | Stmts | Subsystem     | Cover | Stmts  |
|----------------|-------|-------|---------------|-------|--------|
| eli/onboarding | 85%   | 159   | eli/gui       | 45%   | 5,439  |
| eli/world      | 85%   | 971   | eli/contracts | 61%   | 256    |
| eli/coding     | 81%   | 1,045 | eli/planning  | 45%   | 2,132  |
| eli/cognition  | 65%   | 7,021 | eli/execution | 41%   | 12,309 |
| eli/learning   | 62%   | 1,520 | eli/plugins   | 39%   | 1,050  |
| eli/core       | 56%   | 3,492 | eli/system    | 37%   | 261    |

| Subsystem   | Cover | Stmts  | Subsystem         | Cover | Stmts |
|-------------|-------|--------|-------------------|-------|-------|
| eli/memory  | 54%   | 3,338  | eli/tools         | 30%   | 4,508 |
| eli/kernel  | 52%   | 7,059  | eli/perception    | 27%   | 4,138 |
| eli/runtime | 55%   | 13,131 | eli/integrations  | 26%   | 507   |
| api (web)   | 51%   | 1,204  | eli/utils         | 25%   | 468   |
|             |       |        | (gui main window) | excl. | ~7k   |

### Visual (testable surface)



## 5. GUI recovery (the offscreen-Qt effort)

The GUI was 0.6%. The **main window** (eli\_pro\_audio\_gui\_MKI.py, ~7k stmts) blocks on device/display init and can't be built in CI — that stays excluded. Everything else constructs standalone under QT\_QPA\_PLATFORM=offscreen, so lane 4 builds them all for real (27 tests):

| GUI file                 | Cover       | GUI file                         | Cover      |
|--------------------------|-------------|----------------------------------|------------|
| tabs/eli_world_tab.py    | <b>100%</b> | docks/operator_console_dock.py   | <b>67%</b> |
| docks/proactive_dock.py  | <b>94%</b>  | panels/settings.py               | <b>60%</b> |
| tabs/experimental_tab.py | <b>82%</b>  | widgets/ollama_model_selector.py | 50%        |
| panels/startup.py        | <b>61%</b>  | coding_tab.py                    | 49%        |
| labs_tab.py              | 40%         | tabs/tasks_tab.py                | 45%        |

LabsTab builds its full 400+-widget tree; the settings dialog builds 268 children; the startup wizards, both docks, and every tab construct and wire their UI. The only GUI left uncovered is the main window itself (excluded) and residual event-handler branches that need real user interaction.

## 6. Why the remaining low subsystems are low

The low scores cluster around **one cause: the I/O boundaries where ELI touches the real world.** Pure logic is well covered; the edges aren't.

| Subsystem        | Why it's low                                                                                                                                                                                                           |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| execution (41%)  | ~174 action handlers, <b>most side-effecting</b> (open apps, shell, screenshots, media). Can't unit-test "open Firefox" — the test <i>does it</i> . Covered via safe live-lane turns. Router beside it is <b>70%</b> . |
| tools (30%)      | News fetcher (network, gated off), image engine (GPU diffusion), weather (network). Non-network logic is tested; fetch/GPU paths aren't.                                                                               |
| perception (27%) | GPU vision, mic STT, TTS, gaze, live-desktop control — <b>none runs headless</b> . Covered part = the pure parsers (equations 100%, CSV 78%).                                                                          |
| utils (25%)      | Mostly platform_compat.py — if WINDOWS ... elif MACOS ...; on Linux CI only the Linux branch runs. Inherent to cross-platform code.                                                                                    |

## 7. The 5 pre-existing reds (none from this work)

1-3. smart\_home plugin — the in-progress Home-Assistant removal (voice SMART\_HOME now uses ELI's own MQTT server). 4. A blueprint references a since-moved file (eli/execution/handlers/\_\_init\_\_.py). 5. Silent-swallow ratchet — 987 except: pass vs a 950 ceiling (a standing observability debt; the ratchet test correctly forbids raising the ceiling).

## 8. On comparisons (a "closest-in-spirit" project claiming ~89%)

The gap is largely *what ELI is*: a large **embodiment surface** — desktop GUI, gaze/webcam, mic/voice, local vision, OS control, smart-home — that can't fully run in headless CI. A leaner pure-software agent lacks that surface, so a higher fraction of its code is unit-testable by construction. ELI's cognitive **core is in a comparable band** (coding 81%, cognition 65%, kernel 52%), and this report **honestly counts the whole surface** — including the GUI (now 45%) — rather than hiding the hard parts. The residual gap is the genuinely un-automatable edges: side-effecting OS actions, physical hardware, and the one window that can't be built headless.

## 9. Reproduce

```
bash scripts/coverage_full.sh          # runs all 4 lanes, combines, reports
# outputs: artifacts/coverage_report.txt + artifacts/coverage_html/index.html
```

Config: .coveragerc omits only the headless-untestable main window. Every number in this report regenerates from that command on any checkout.